Literature Review, Systematic Experimentation and Extension of Masked Autoencoders are Scalable Vision Learners

1. Literature Review

As we already know Transformer (2017) architecture in class, in Natural Language Processing (NLP), the paradigm was revolutionized by BERT (Devlin et al., 2018), which introduced Masked Language Modeling (MLM). By masking a small portion of the input text and training the model to predict the missing words, BERT demonstrated the immense power of autoencoding for pre-training.

However, translating this success to Computer Vision (CV) was initially challenging due to architectural differences and the dense, highly redundant nature of image pixels. The architectural barrier was fundamentally lowered by the Vision Transformer (ViT) (Dosovitskiy et al., 2020). By dividing an image into a sequence of non-overlapping patches and processing them with a standard Transformer, ViT provided a sequence-based representation of images, making NLP-like self-supervised tasks structurally feasible in vision.

Prior to Masked Image Modeling (MIM), CV Self-supervised learning (SSL) was heavily dominated by contrastive learning methods that we have already learned in Lecture 10. While effective, these methods relied heavily on carefully engineered, complex data augmentation pipelines to create positive and negative sample pairs. The focus article, Masked Autoencoders (MAE), represents a paradigm shift away from contrastive learning towards generative MIM. As highlighted in the comprehensive survey by Zhang et al. (2022), MAE established a foundational framework that successfully adapted the MLM philosophy to vision, proving that simple reconstruction tasks could yield highly transferrable representations.

Contributions and Improvements over Existing Methods

- Asymmetric Encoder-Decoder Architecture for Scalability. The most significant architectural contribution of MAE is the decoupling of the encoder and decoder. A lightweight decoder then processes the encoder's latent representations alongside mask tokens to reconstruct the original image.
- High Masking Ratio Addressing Information Redundancy. Unlike BERT in NLP, which typically masks only 15% of tokens due to the high information density of language, MAE employs an aggressively high masking ratio of 75%. The authors identified that images are highly redundant; if the masking ratio is too low, the model can solve the task through simple low-level pixel interpolation.
- Simplification of the Pre-training Pipeline. Before MAE, state-of-the-art CV SSL required complex data augmentations, massive batch sizes, and memory banks. MAE completely abandons these requirements. It relies on a straightforward generative task—reconstructing raw pixel values using a simple Mean Squared Error (MSE) loss on the masked patches.

The profound impact of MAE has inspired numerous extensions. For instance, while the original MAE relies on random patch masking, subsequent works like SemMAE (Li et al., 2022) introduced semantic-guided masking strategies.

2. Understanding and Visualization of MAE: From Sequence Processing to Masked Image Modeling

The architecture follows an interesting route: from the original Transformer to the ViT, and finally to the MAE framework.

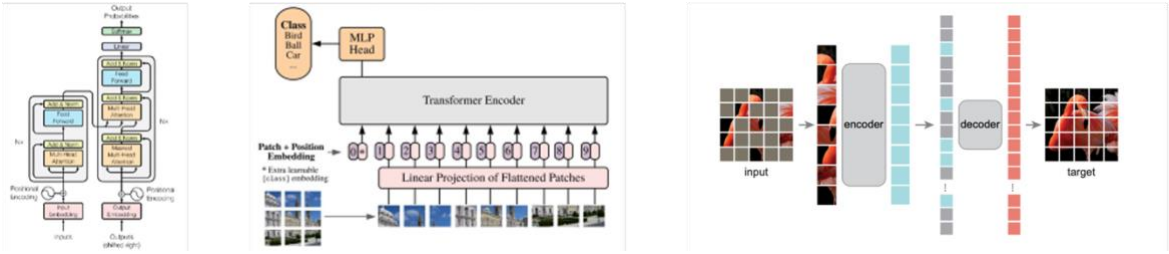

Figure 1 Architecture of Transformer, Vision Transformer and MAE

2.1. The Conceptual Bridge: From Transformers to ViT

The Transformer architecture was originally designed for NLP to process sequential data (words) using a self-attention mechanism, which excels at capturing long-range global dependencies. However, translating this to images was historically difficult because images are grids of pixels, not 1D sequences.

The ViT bridged this gap by treating an image exactly like a sentence. An image of size $H \times W \times C$ is divided into a grid of non-overlapping patches (16×16 pixels per patch in paper example). These patches are linearly projected into 1D vectors called "Patch Embeddings". Because the self-attention mechanism is permutation-invariant, Positional Embeddings are added to these patches to retain spatial information. This flattened sequence of patches is then fed into a standard Transformer block.

2.2. The MAE Architecture: Asymmetric Autoencoding

Building upon the ViT backbone, the focus article introduces MAE. The methodology operates through a highly optimized, asymmetric encode-decode process:

Step 1: Aggressive Random Masking. The input image is divided into patches following the ViT approach. A remarkably high proportion—typically 75%—of these patches are randomly masked out. This high masking ratio is a critical working detail: unlike text, images contain heavy spatial redundancy. If only a few patches are masked, the model can "cheat" by interpolating nearby pixel values. Masking 75% forces the model to learn deep, holistic semantic representations of the objects to reconstruct the missing parts.

Step 2: The Heavyweight Encoder. The MAE Encoder is a standard, large-scale ViT. However, it operates strictly on the 25% visible, unmasked patches. This is the core engine of MAE's scalability. By ignoring the masked tokens entirely, the computation and memory footprint are reduced to a fraction of a standard ViT, allowing for the training of massive models efficiently. The encoder maps these visible patches into rich latent representations.

Step 3: The Lightweight Decoder. The decoder's job is to reconstruct the original image. Its input consists of the encoded visible patches and Mask Tokens.

Step 4: Additional Details. A mask token is a shared, learnable vector that indicates the presence of a missing patch to be predicted. Crucially, full positional embeddings are added to all tokens in this decoder phase. Without this, the mask tokens would have no information about their original location in the image grid. Since the decoder only performs the reconstruction task during pre-training, it can be designed to be much narrower and shallower than the encoder, further significantly reducing computation time.

2.3. Reconstruction Target and Loss Function

The output of the decoder is a tensor representing the predicted pixel values for each patch. The loss function computes the Mean Squared Error (MSE) between the reconstructed pixels and the original image pixels. A vital methodological detail is that the loss is calculated only on the masked patches, identical to BERT's logic. Computing loss on the visible patches would encourage the model to waste capacity memorizing the input rather than learning to predict the unseen.

2.4. Examples

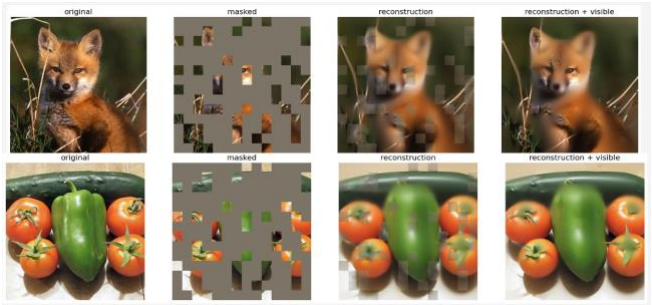

Figure 2 Examples of Masked and Reconstructed Images

From the code demo provided by the paper team in GitHub, we can run and output two examples as listed above.

- Original: 100% real picture, the whole picture.
- Masked: the visible part for encoders to input, which means the model uses 25% to predict the remaining 75% part.
- Reconstruction: the output from the decoders.
- Reconstruction + Visible: 25% part used in encoder + 75% part from the decoder.

Using the pre-trained model provided in GitHub, we can see clearly that the method is very powerful. The model is well trained and the prediction accuracy for the unmasked parts is very high.

3. Systematic Experimentation on the Hyperparameters on MAE

We reproduce the Masked Autoencoder (MAE) framework (He et al., 2021) on CIFAR-10 (32×32), experimenting across three axes: masking ratio, patch size, and encoder capacity. Our implementation follows the original asymmetric encoder-decoder design — a Vision Transformer (ViT) encoder (Dosovitskiy et al., 2020) operating only on visible patches, with a lightweight decoder reconstructing masked patches using per-patch normalized MSE loss. Training uses AdamW (β_1=0.9, β_2=0.95) with cosine LR decay and warmup, directly mirroring He et al. (2021). The approach is conceptually rooted in BERT-style masked language modelling (Devlin et al., 2018), adapted from text to vision: randomly mask tokens and train the model to reconstruct them from context alone. The key differences from He et al. (2021) are scale — we use CIFAR-10 vs ImageNet-1K, 50 epochs vs 1600, and models up to 20M parameters vs 632M.

3.1 Experiment 1 – Masking Ratio

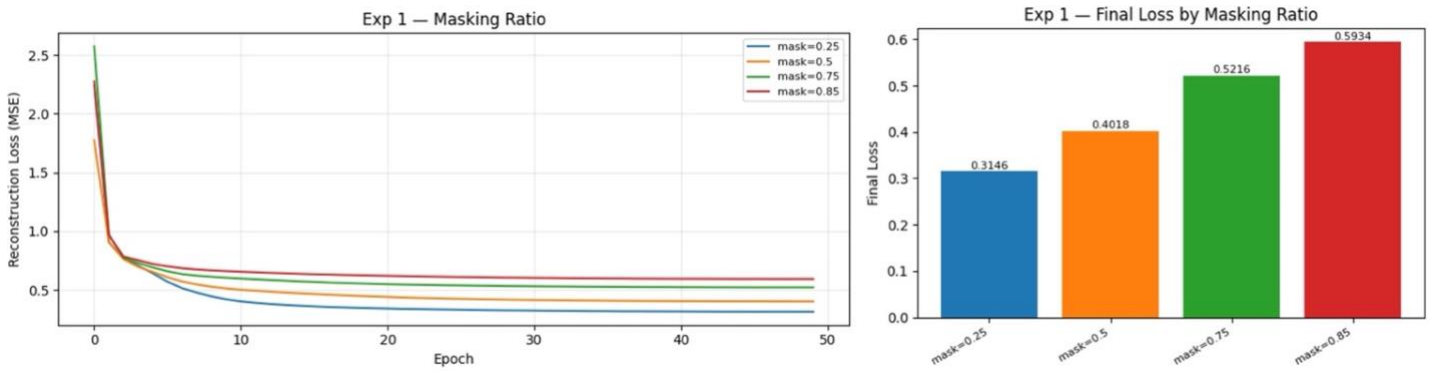

Figure 3 Reconstruction Loss in varying Masking Ratio

Masking ratios of 0.25, 0.50, 0.75, and 0.85 were tested with patch=4×4 and encoder=Small. Final losses were 0.3146, 0.4018, 0.5216, and 0.5934 respectively — monotonically increasing with masking ratio, exactly as He et al. (2021) predicts. Lower masking means more visible context, making reconstruction easier and MSE lower. However, mask=0.25 is not optimal for representation learning: the task is too easy, encouraging texture interpolation rather than semantic understanding. He et al. (2021) recommends 0.75 specifically because it forces the model to reason about global structure from sparse context — a harder task that produces richer learned representations.

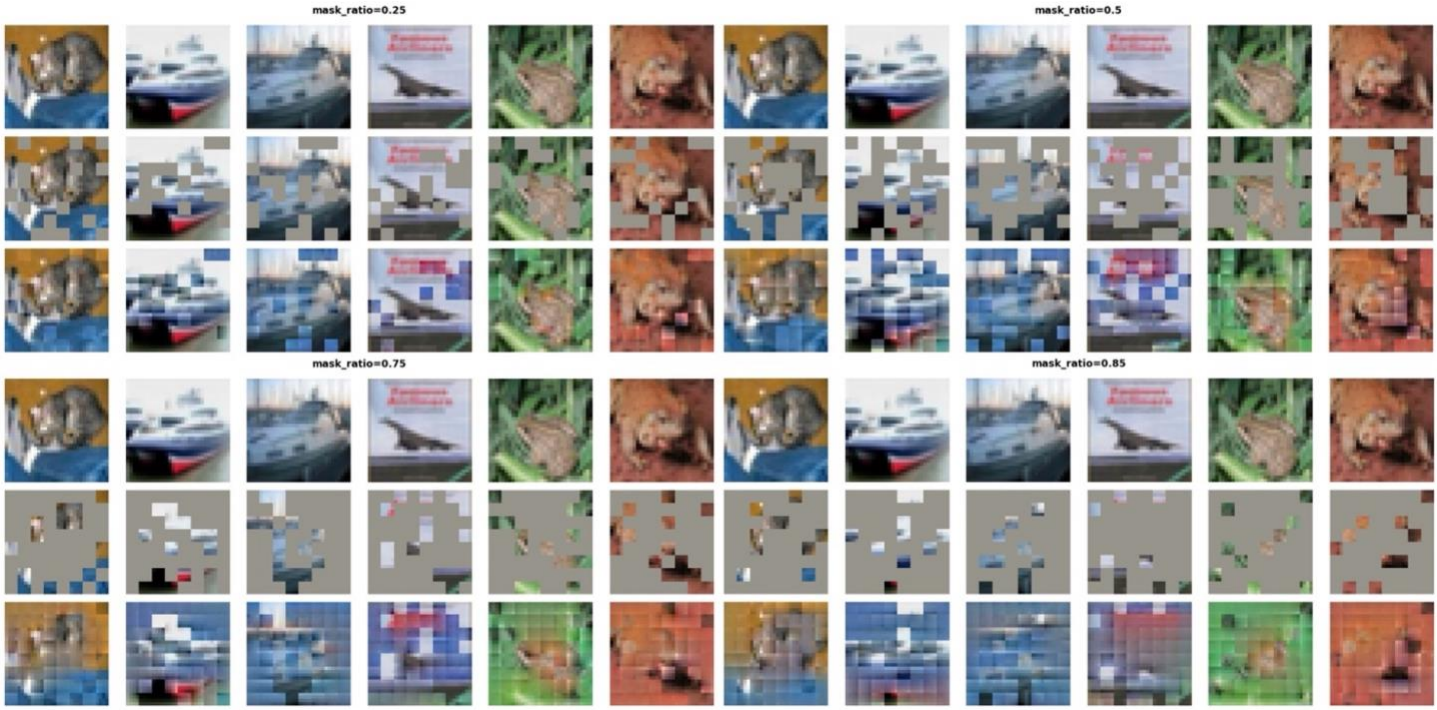

Figure 4 Reconstructed Images for different Masking Ratio

3.2 Experiment 2 – Patch Size

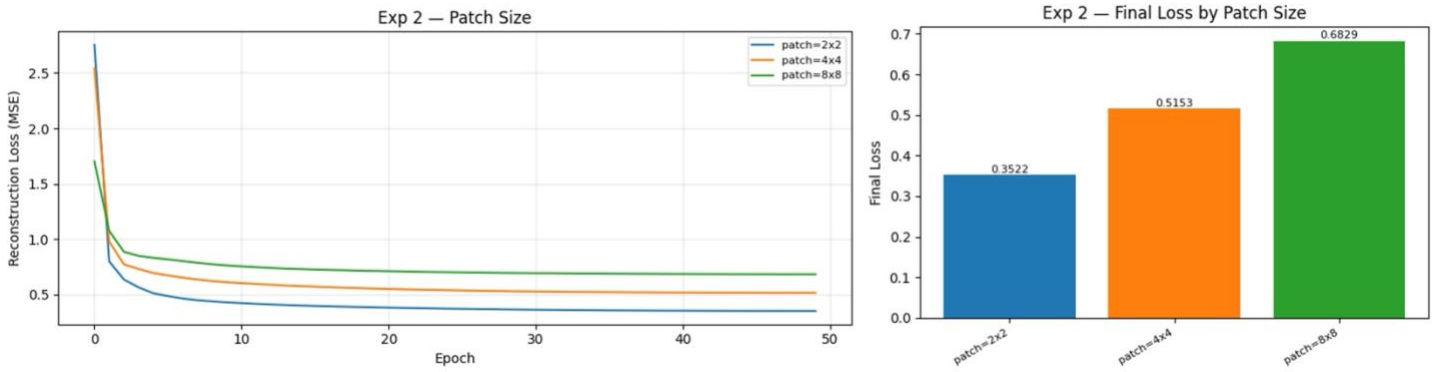

Figure 5 Reconstruction Loss in varying Patch Size

Patch sizes of 2×2 (256 tokens), 4×4 (64 tokens), and 8×8 (16 tokens) were tested at mask=0.75. Final losses were 0.3522, 0.5153, and 0.6829. Patch=2×2 achieves the lowest loss because fine-grained tokens are individually easy to reconstruct, and abundant visible context remains even at 75% masking. Patch=8×8 is the worst — only 4 patches remain visible at 75% masking on a 32×32 image, leaving the decoder with almost no spatial information. Despite its low loss, Patch=2×2 is not theory-optimal: Patch=4×4 (64 tokens) is proportionally equivalent to the 16×16 patches used by He et al. (2021) on 224×224 images, representing the best balance between task difficulty and spatial context for a 32×32 image. Consistent with Dosovitskiy et al. (2020), patch granularity fundamentally governs the richness of the token sequence fed to the transformer encoder.

Figure 6 Reconstructed Images for different Patch Sizes

3.3 Experiment 3 – Encoder Size

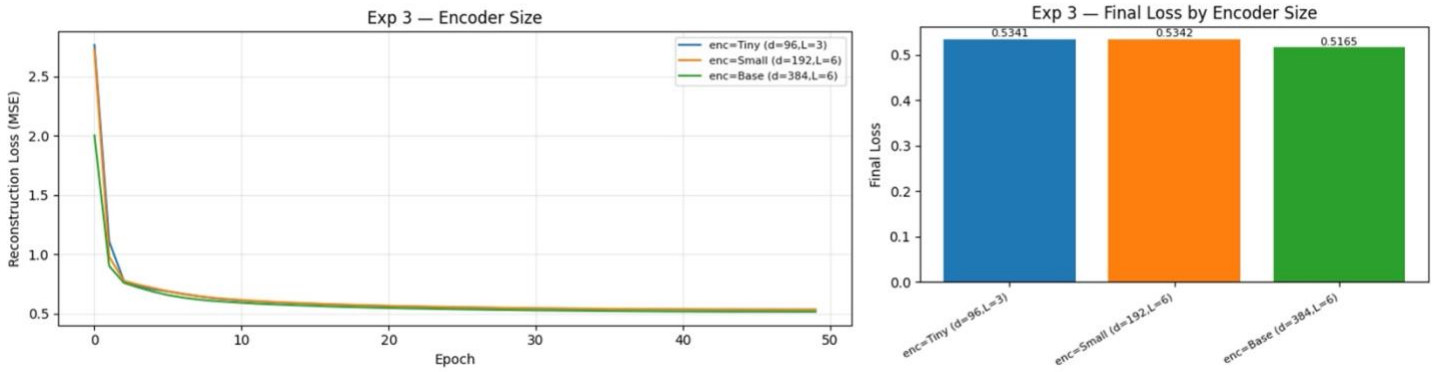

Figure 7 Reconstruction Loss by varying Encoder Size

Tiny (96d, 3L), Small (192d, 6L), and Base (384d, 6L) encoders were tested at mask=0.75, patch=4×4. Final losses were 0.5341, 0.5342, and 0.5165. Tiny and Small are virtually identical (Δ=0.0001), while Base achieves a modest 3.3% improvement. The near-zero gap between Tiny and Small reflects a data-limited regime: CIFAR-10's low resolution and 50 epochs of training do not provide sufficient learning signal to differentiate encoder capacities. Base's advantage confirms that capacity matters, but the scalability claim of He et al. (2021) — that larger models benefit disproportionately from MAE pre-training — fully manifests only with longer training on harder, higher-resolution data.

Figure 8 Reconstructed Images by different Encoder Sizes

3.4 Conclusion

The loss-optimal configuration is mask=0.25, patch=2×2, encoder=Base — but this is not the recommended configuration. The theory-optimal setting is mask=0.75, patch=4×4, encoder=Base, directly mirroring the validated choices of He et al. (2021). The central takeaway is that reconstruction loss and representation quality are not the same thing (He et al., 2021): the harder the reconstruction task, the richer the learned visual representations, even at the cost of higher MSE. Future work should add linear probing evaluation to empirically confirm this on downstream CIFAR-10 classification.

4. Masked Autoencoders on Medical Images

In the medical field, obtaining high-quality, annotated datasets is a significant setback. Re-imaging patients' pictures (e.g., CT/MRI) due to blurriness or artificial organs is often unfeasible due to cost and radiation risks. Masked Autoencoders (MAE) address this problem by learning the underlying structure of medical data through reconstruction, effectively "filling in the blanks" when data is incomplete or unlabeled (He et al., 2022).

When we tried to run MAE on medical (VQA-RAD) images (Lau et al., 2018), we achieved a Mean Squared Error (MSE) of 0.1716, which translates to a Reconstruction Fidelity of 82.84%. Despite a high masking ratio of 75%, the model demonstrated a robust ability to infer missing anatomical structures, consistent with the Vision Transformer (ViT) architecture's scalability (Dosovitskiy et al., 2020). Furthermore, the standard deviation of 0.1099 indicates that the model's performance remains consistent across diverse radiological samples, proving its potential as a reliable tool for enhancing or reconstructing low-quality medical imagery without additional radiation exposure.

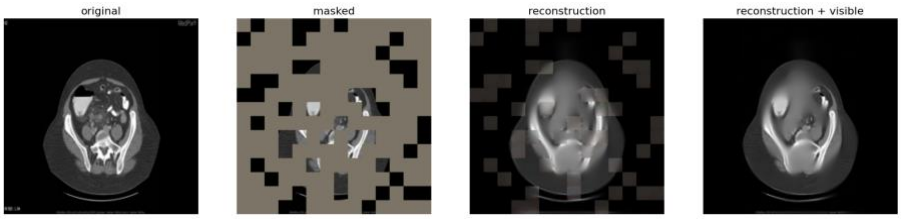

Figure 9 MAE on Medical Images

Secondary Studies about MAE on medical field

According to the study on Multi-Modal Masked Autoencoders (Chen et al., 2022), medical diagnosis is rarely based on images alone; it involves a synergy between vision (X-rays/CT) and language (radiology reports). MAE facilitates Vision-and-Language Pre-training (VLP) by masking both image patches and text tokens. This allows the model to learn cross-modal correlations, helping the AI understand how specific visual patterns in an MRI relate to technical descriptions in a clinical report (Chen et al., 2022).

Also, while MAE may provide powerful representations, practical deployment requires efficiency. Insights from lightweight and hierarchical models suggest that hierarchical optimization can refine the reconstruction process. In a medical context, this means the model can focus on "reshaping" critical anatomical features with high precision while remaining computationally efficient enough for real-time hospital use.

Overall, MAE shifts the paradigm from requiring massive, labeled datasets to leveraging the inherent structure of medical images and text. By reconstructing masked information, these models provide a robust foundation for more accurate, multimodal, and efficient diagnostic tools.

5. Semantic Masking in Autoencoders

This study modifies the standard Masked Autoencoder (MAE) (He et al., 2022) by replacing uniform random masking with semantically guided masking based on CLS to patch attention maps from a supervised teacher ViT (Dosovitskiy et al., 2021). Three

variants were pretrained on CIFAR-10 (Krizhevsky, 2009) under identical settings random masking, semantic-high (masking semantically important foreground patches), and semantic-low (masking less important background patches) using the same ViT-Tiny encoder, 2-layer decoder, AdamW optimiser, 75% mask ratio, and 50 pretraining epochs. The only difference across models was how the 48 masked patches were chosen.

Figure 10 Teacher CLS-to-patch attention (bottom row) confirms foreground bias even at 70% accuracy.

The main result is that semantically guided masking produces stronger and more transferable representations than random masking. Although semantic-high had the highest reconstruction loss (0.598), it achieved the best linear-probe accuracy (46.76%), while semantic-low reached 46.41%, both clearly outperforming the random baseline (41.05%). This shows that lower reconstruction loss does not necessarily imply better representations; instead, harder, semantically targeted reconstruction encourages the encoder to learn richer features. The semantic variants also converged faster during probing, with semantic-high reaching 39.7% test accuracy in the first probe epoch compared with 30.4% for random, indicating better linear separability from the start.

5.1 Quantitative Results

Table 1 Reconstruction loss and linear probe accuracy (20-epoch frozen-encoder probe). Blue = improvement over baseline

Strategy	Recon. Loss (ep 1)	Recon. Loss (ep 50)	Best Probe Acc	Gain vs Random
Random (baseline)	1.042	0.457	41.05%	—
Semantic-High	1.073	0.598	46.76%	+5.71 pp
Semantic-Low	1.004	0.582	46.41%	+5.36 pp

Following masking pattern based on semantic masking, the figure below compares the three masking strategies. Random masking distributes grey-masked patches uniformly. Semantic-high preferentially masks the object, leaving mostly background visible. Semantic-low does the inverse, masking background and leaving the object.

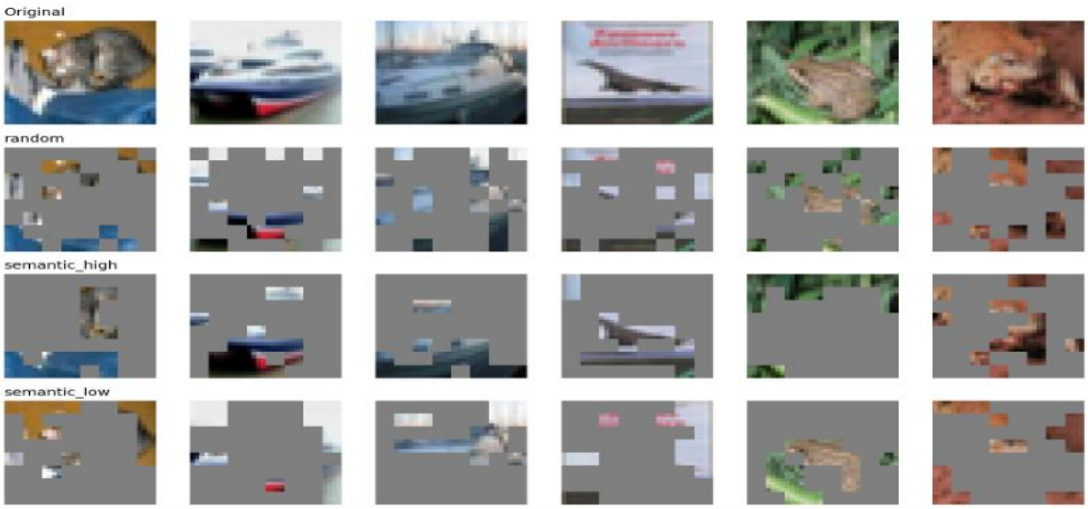

Figure 11 Masking patterns. Random distributes uniformly; semantic-high masks foreground; semantic-low masks background

All three decoders produce plausible blurred reconstructions in the masked regions, confirming that no variant's decoder is broken. The semantic-high reconstructions are notably harder, yet the encoder learns better representations despite higher reconstruction loss. Semantic masking improves representation quality. Both variants beat random by ~+5.5 pp in linear-probe accuracy (46.76% and 46.41% vs 41.05%), a substantial gain at this scale with no architectural cost. These results raise the question of why both semantic variants outperform random masking, regardless of whether foreground or background is masked.

5.2 Discussion

Content-aware structure, not mask target, drives improvement. The near-parity between masking foreground (semantic-high) and background (semantic-low) shows the benefit comes from structured, content-aware mask selection rather than any specific mask-target relationship. Reconstruction loss and feature quality are decorrelated. The strategy with the worst reconstruction loss (semantic-high, 0.598) learns the best features. Harder reconstruction forces richer encoding, consistent with the broader MAE literature (Zhang et al., 2022). The masking strategy is a high-leverage design axis. Alongside patch size (identified in baseline experiments), the masking policy is one of the most impactful choices for MAEs on low-resolution imagery.

5.3 Limitations

This is a deliberately small-scale controlled comparison (ViT-Tiny, 32×32, 50 epochs, frozen probe). Absolute accuracy is modest (~47%). Whether the ~5.5 pp gap persists at larger scales, higher resolutions, or with full fine-tuning remains open. The semantic signal also requires a supervised teacher, introducing label dependency that pure self-supervised methods avoid. If confirmed at scale, these findings would further validate the masking strategy as a first-class design axis for masked autoencoders.

6. Conclusion

In this report, we conducted an in-depth investigation into Masked Autoencoders (MAE), covering their fundamental operations, architectural design, and practical efficacy. By utilizing a pre-trained model provided on GitHub, we observed that the MAE framework is exceptionally powerful; the model is well-trained, and its prediction accuracy for unmasked regions remains remarkably high.

Our experimental analysis revealed a critical distinction between reconstruction performance and representation quality. While the loss-optimal configuration was found at a 25% masking ratio with a 2×2 patch size and a Base encoder, this is not the recommended setting for downstream tasks. Instead, following He et al. (2021), we advocate for the theory-optimal configuration — a 75% masking ratio and 4×4 patch size — which forces the model to reason about global structure from sparse context rather than interpolating local texture. The central takeaway is that reconstruction loss and representation quality are not the same thing: the harder the reconstruction task, the richer the learned visual representations are expected to be, even at the cost of higher MSE (final loss of 0.5216 at mask=0.75).

Furthermore, our findings suggest that improvement is driven by content-aware structure rather than specific mask targets. The near-parity between masking foreground (high-semantic) and background (low-semantic) regions indicates that the benefit arises from structured, content-aware selection. Alongside patch size, the masking policy emerges as one of the most impactful design axes for MAEs, particularly on low-resolution imagery. Harder reconstruction tasks force the encoder to capture more complex latent features, consistent with broader MAE literature.

Overall, MAE shifts the paradigm from requiring massive, labeled datasets to leveraging the inherent structure of images and text. This was observed when assessing medical images, where image itself is not enough for users. By reconstructing masked information, these models provide a robust foundation for more accurate, multimodal, and efficient diagnostic tools. Future work should incorporate linear probing evaluations to empirically confirm these findings on downstream tasks, such as CIFAR-10 classification.

7. References

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8. Contributions

Name	Contributions
Gao Demin	1. Explanation for Methodologies of MAE 2. Developed code for visualization of MAE 3. Prepared Content for Literature Review and visualization of MAE 4. Contributed to brainstorming and organizing the structure and content of the report.
Kong Moon Hyeong	1. Developed code for MAE on Medical Images 2. Prepared Content for MAE on Medical Images 3. Contributed to brainstorming and organizing the structure and content of the report. 4. Contributed to the overall conclusion
Mahapatra Onkar	1. Developed code for Semantic Masking on MAE 2. Prepared Content for Semantic Masking on MAE and the limitations based on experimental setup. 3. Analyzed the experimental results, identifying key findings such as the decorrelation between reconstruction loss and representation quality. 4. Contributed to brainstorming and organizing the structure and content of the report.
Sivakumar Nishaanth	1. Developed code for experimentation on MAE by varying hyperparameters 2. Prepared Content for experimentation on MAE by analyzing the results from the developed code and comparing it with the published papers and theory 3. Contributed to brainstorming and organizing the structure and content of the report. 4. Created and maintained the overall structure and organization of the GitHub repository.

All codes and results associated with this work are available in the GitHub repository - https://github.com/NishaanthSivakumar/ST5229-Deep-Learning-Project